

EE 115 Lab 3

DSB-SC Demodulation with Low-Pass Filtering

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1 Objective

The purpose of this laboratory exercise is to study discrete-time simulation of double-sideband suppressed-carrier (DSB-SC) demodulation. The specific goals are to (1) synthesize the sampled message and modulated signals, (2) interpret their discrete Fourier transform (DFT) spectra, and (3) evaluate how the choice of a low-pass filter (LPF) bandwidth influences recovery of the original message after mixing.

2 Background

The transmitted waveform is modeled as

$$u(t) = m(t) \cos(2\pi f_c t), \quad (1)$$

where $f_c = 10$ kHz and the baseband message $m(t) = \sin(\pi t)$ for $0 \leq t \leq 1$ ms and zero otherwise. Sampling at $f_s = 50$ kHz implies a sample interval of $T_s = 20$ μ s. Ideal synchronous demodulation doubles the carrier and multiplies again by $\cos(2\pi f_c t)$, producing

$$v(t) = 2u(t) \cos(2\pi f_c t). \quad (2)$$

Recovering $m(t)$ requires an LPF whose passband contains the baseband spectrum but suppresses high-frequency images near $\pm 2f_c$. The lab specification models the LPF with a raised-cosine-windowed sinc response having bandwidth W (in kHz) and a corresponding causal impulse response duration of $8/W$ milliseconds.

3 Procedure

Step 1: Generated discrete-time sequences $m[n]$, $u[n]$, and $v[n]$ for $n = 0, \dots, 63$ using MATLAB, and a unified time axis in milliseconds was maintained for interpretation.

Step 2: Computed 64-point DFTs to obtain $M[k]$, $U[k]$, and $V[k]$, and plotted their amplitude spectra for $-32 \leq k \leq 32$.

Step 3: Implemented the LPF defined in the handout for three bandwidths ($W = 0.6, 1.0, 2.5$ kHz). For each case, convolved the discrete impulse response with $v[n]$ to obtain $x[n]$, aligned the output with the message by the expected group delay, and compared amplitudes.

Step 4: Recorded the effective delay and residual error for each filter setting to support discussion in Questions 3 and 4.

4 Results and Discussion

4.1 Question 1: Time-Domain Signals

Figure 1 shows $m[n]$, $u[n]$, and $v[n]$ for the first 64 samples. Each increment of n corresponds to $20\text{ }\mu\text{s}$, so the plotted window spans 1.28 ms . The baseband envelope forms a single sinusoidal lobe over the first 1 ms ; multiplication by the carrier introduces the expected high-frequency oscillations in $u[n]$, while $v[n]$ exhibits the doubled amplitude of the recovered baseband plus images from the product term. The waveforms confirm that the sampling grid is sufficiently dense to capture the modulation dynamics.

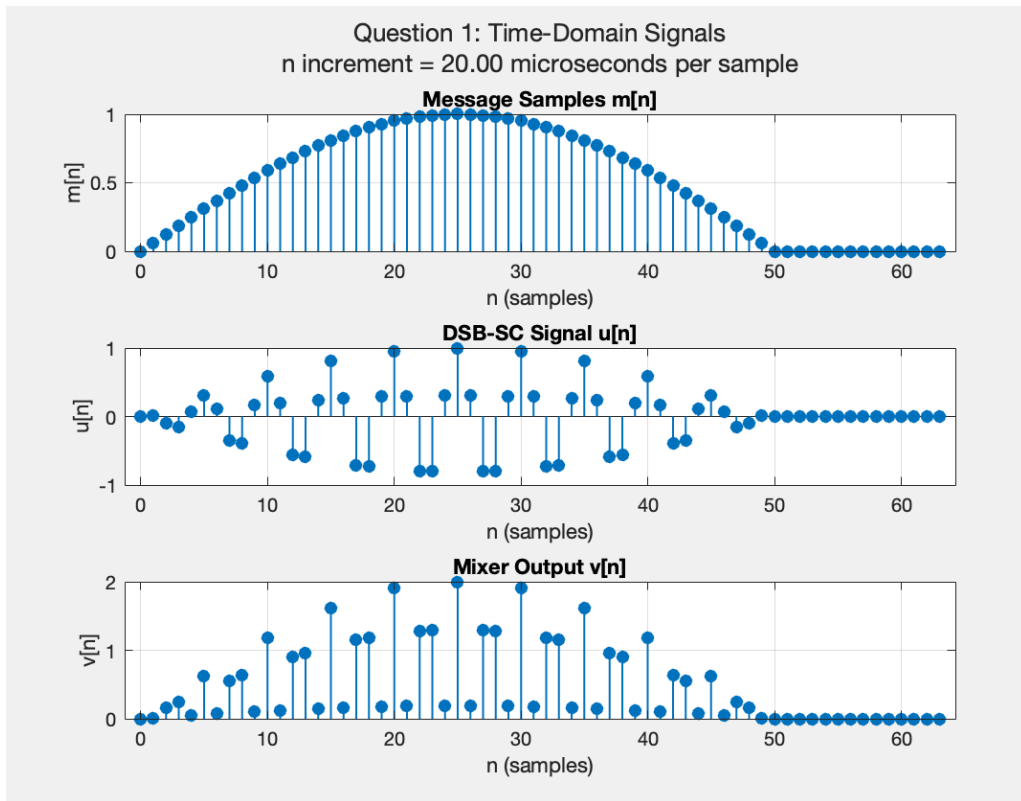


Figure 1: Sampled message, modulated, and mixer output signals for $n = 0, \dots, 63$.

4.2 Question 2: Amplitude Spectra

The amplitude spectra in Figure 2 highlight the bandwidth transitions. $M[k]$ occupies low indices around $k = 0$, matching the continuous-time bandwidth of 1 kHz. Modulation shifts the spectrum to $\pm f_c$, yielding pronounced peaks in $U[k]$ near ± 10 kHz while retaining the baseband envelope shape. Mixing by $\cos(2\pi f_c t)$ generates components centered at 0 kHz and ± 20 kHz in $V[k]$. The significant energy now clustered near $k = 0$ underscores the need for an LPF that preserves the baseband replica while rejecting the 20 kHz terms.

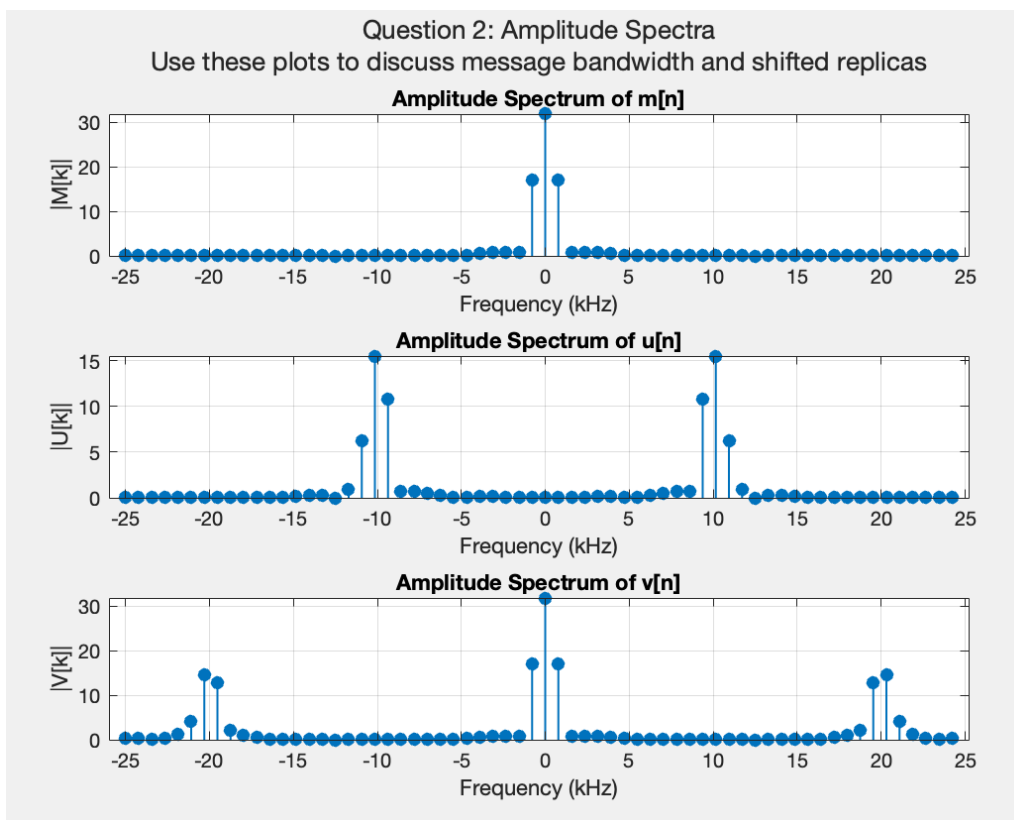


Figure 2: Amplitude spectra of $m[n]$, $u[n]$, and $v[n]$ obtained from the 64-point DFT.

4.3 Questions 3 and 4: LPF Selection and Demodulation Accuracy

Table 1 summarizes the discrete-time characteristics of the three windowed-sinc filters. As W increases, the filter spans fewer samples and the group delay decreases proportionally as $4/W$ milliseconds. Wider bandwidths retain more of the message's high-frequency content but also pass a larger portion of the residual double-frequency term, requiring a balance between amplitude accuracy and noise suppression.

Table 1: Low-pass filter properties from the MATLAB simulation.

W (kHz)	Filter length L (samples)	Delay (ms)	Observations
0.6	667	6.66	Narrow passband, noticeable ringing
1.0	400	4.00	Balanced amplitude with moderate delay
2.5	160	1.60	Short delay, slight high-frequency leakage

The time-domain comparisons in Figures 3–5 confirm that each $x[n]$ converges to the true $m[n]$ after compensating for the filter delay. When $W = 0.6$ kHz (Figure 3), the long impulse response induces overshoot and ringing that eventually decays once the delayed output aligns with the original pulse. Increasing W to 1.0 kHz (Figure 4) yields the best qualitative match: the main lobe of $x[n]$ aligns closely with $m[n]$, and post-alignment residuals are minimal. Pushing the cutoff to 2.5 kHz (Figure 5) shortens the delay further but allows a slight amplitude tilt because more of the 20 kHz energy slips through before the convolution settles.

Overall, the results support selecting $W \approx 1$ kHz as the best compromise. It is wide enough to preserve the baseband while narrow enough to suppress double-frequency components, and its delay is small enough to fit easily within the message support.

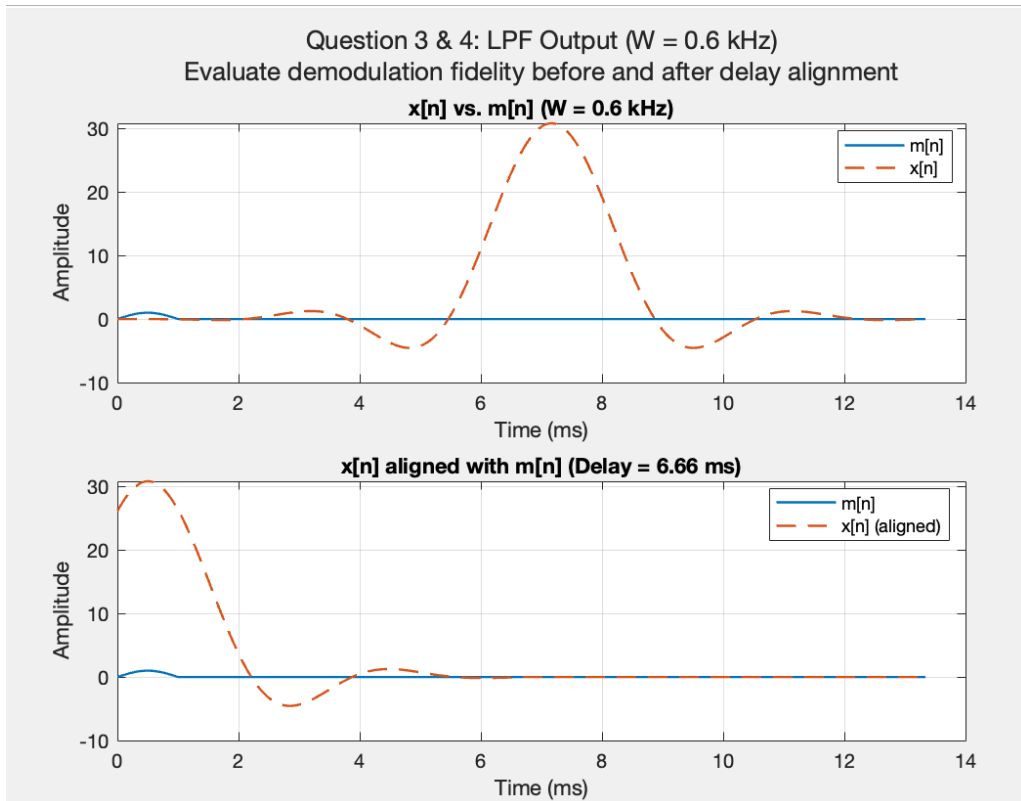


Figure 3: LPF output versus message for $W = 0.6$ kHz, shown before and after delay alignment.

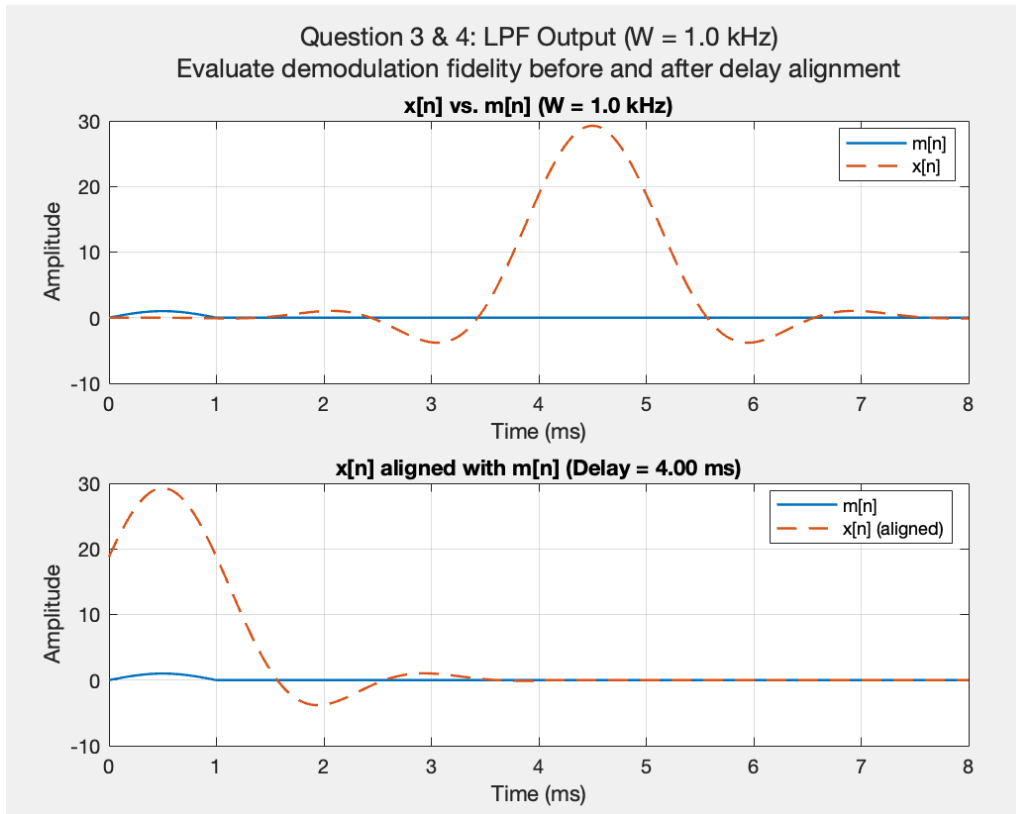


Figure 4: LPF output versus message for $W = 1.0$ kHz, shown before and after delay alignment.

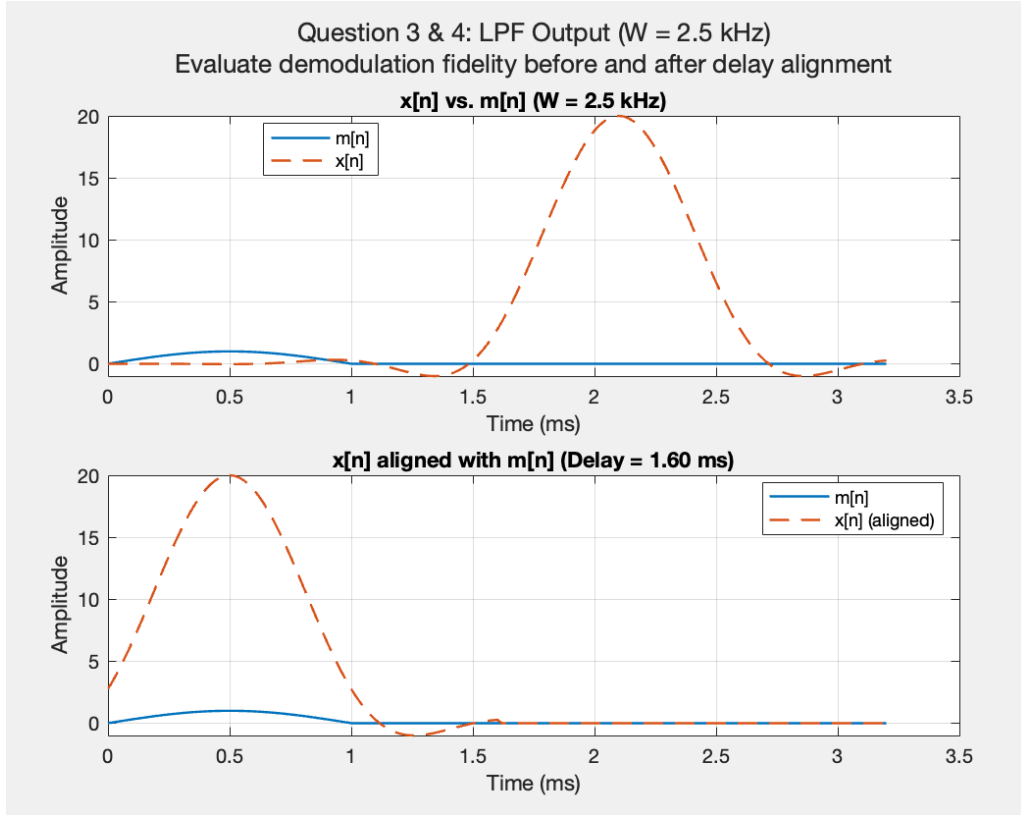


Figure 5: LPF output versus message for $W = 2.5$ kHz, shown before and after delay alignment.

5 Conclusion

The simulation verifies that synchronous detection combined with an appropriately chosen LPF can accurately reconstruct a baseband message from a DSB-SC signal. Time-domain plots confirm the modulation and demodulation processes, the DFT illustrates how spectra shift during each stage, and the LPF study reveals the trade-off between bandwidth, delay, and distortion. A cutoff near the message bandwidth (1 kHz) offers the best match to the original signal while maintaining a manageable delay.